Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/14/2021
PI Name:	Liu, Bin Ph.D.		
Project Title:	Three-Dimensional Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	07/25/2017	**End Date:**	10/26/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/26/2021 and contract number changed to JPL 1661767 per U. Israelsson/JPL (Ed., 5/24/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goree, John Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1661767; JPL 1579454		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A flight experiment will be performed to study phenomena resulting from shear motion in a dusty plasma. Applying an external shear, we will probe fundamental material properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. We will take a microscopic approach to uncover the nature of macroscopic properties. We will use mesoscopic particles, which serve as analogs of molecules, but allow tracking of individual particles as they interact. The experiment will use dusty plasmas, also known as "complex plasmas." These are four-component mixtures of micron-size solid particles, electrons, ions, and neutral gas. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. Microgravity enables our experiment with a three-dimensional sample. Without microgravity, under 1g conditions, sedimentation causes the dust cloud to settle into a 2D layer, or at most a few horizontal layers. Our microgravity experiment will extend the understanding of particle motion that was gained in previous ground-based experiments with smaller 2D samples. Our flight experiment will be performed using Plasma Kristall–4 (PK-4), which is a facility in the Columbus module of the International Space Station (ISS). It was built by the European Space Agency. The main diagnostic is video imaging using cameras. Image data will be analyzed to generate particle position and velocity. Physical measures that can be obtained from these image data, using methods developed for ground-based experiments, include a wave's propagation speed and dispersion relation, temperature profile, flow velocity, or shear stress. SCIENTIFIC/TECHNICAL MERITS The experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? The experiment will exploit manipulation devices on PK-4 including laser manipulation. Laser light imparts momentum to the solid particles, causing them to flow through the surrounding sample region. This modulation will be a pulse that is either long or short, according to the scientific goal. The feasibility of our flight experiment is based on previous ground-based experiments and our simulation. Our laboratory experiments have confirmed localized viscous heating and lattice waves. Ground-based experiments by other scientists have also confirmed the existences of plastic flow conditions and stick-slip motion. Using a simulation under PK-4 conditions, we have established the physical mechanism of wave excitation by a pulsed shear flow. In the unlikely event that this flight experiment does not yield useful results, we have a plan for a backup experiment to investigate the superheating and supercooling phase transitions, or non-Gaussian statistics. BROADER IMPACTS This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
	In this report period, significant progress has been made in analyzing the experimental data from our PK-4 flight experiment executed in Science Campaign 7 on the International Space Station (ISS). Particularly, our particle image velocimetry (PIV) analysis revealed the onset of an unsteady shear motion in a dusty plasma, advancing our understanding on how strongly-coupled plasmas respond to impulsive forces at small time scales. This plasma contains a collection of micron-size dust particles immersed in an ionized neon gas. These particles obtained large electric charges by absorbing electrons and ions in the plasma, constituting a strongly-coupled plasma (also called a complex plasma) in three dimensions under microgravity conditions. In addition, we also made progress in particle tracking, new analysis method development, as well as structural and dynamical characterization of the 3D dusty plasmas. TIME-DEPENDENT SHEAR MOTION IN STRONGLY-COUPLED DUSTY PLASMAS Our research effort has led to a new manuscript for publication. This manuscript, titled "Time-Dependent Shear Motion in a Strongly-Coupled Dusty Plasma in PK-4 on the International Space Station (ISS)," was recently submitted to IEEE Transactions on Plasma Science. [Ed. Note: Bibliography has been updated to include this paper.] Several highlights from this paper include: (1) We reported an observation of impulsive shear flows in a 3D strongly-coupled plasma. (2) At the onset of the impulsive motion, we observed the flow growing in space and time. Later, during a steady period, we find the thickness of the flow region fluctuated significantly over time. (3) Our research approach was essentially microscopic because our dust fluid velocity was evaluated at a length scale of a few particle spacings. Scientific knowledge we gained from this research includes: • In a dust cloud in PK-4 under laser manipulation, we observed three distinctive structural domains, including a region of rapid flow inside the laser beam, a transitional shear layer near the edge of the beam, and a nearly undisturbed domain far from the beam. • The dust flow driven by a manipulation laser in PK-4 exhibited significant fluctuations. These fluctuations, which were not previously investigated, exhibited different characteristics in different domains, indicating the existence of different physics mechanisms, for example, those arising from a local particle-particle collision or from macroscopic effects at

	hydrodynamic scales.
Task Progress:	• We found that our dust cloud exhibited motion transverse to the flow made by the manipulation laser. This suggests a possible role for viscous heating, which can scatter flow energy to the transverse direction. • We found the growth of the dust flow upon an impulsive force followed a trend that can be explained by neutral gas friction. PARTCLE IMAGE VELOCIMETRY (PIV) To evaluate the flow velocity in our strongly-coupled dusty plasma, we developed our own computer software for particle image velocimetry (PIV) method. A benefit from in-house developed software is that it can be easily tailored to our needs more accurately. We implemented our PIV analysis in three main steps, including preprocessing images, evaluating particle displacement using a direct cross correlation of the images from two consecutive video frames, and validating the PIV data. This code has been used to our analysis of the shear flows in this project and for velocity field in dust acoustic waves for another Iowa project. TRACKING DUST PARTICLES IN A FLOW We developed a new method to track particles in a flow. For our flight experiment, the dust flow had a velocity of about 10 mm/s, which is much larger than typical dust thermal velocity in a strongly-coupled plasma. Consequently, our previous particle-tracking method is unable to thread the particles in the flow. To overcome this problem, we devised a scheme that considers of the mean velocity of the particle flow. We found that our new method was able to track more than 90% of the particles in the flow. COLLABORATION WITH OTHER SCIENTISTS We worked closely with PK-4 Core Team members in Germany and Russia. In general, our research requires close collaboration with several other institutions or agencies, including the PK-4 Core Team members, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Liu B, Goree J. "ISS flight experiments measuring acoustic waves and microscopic flows in three-dimensional dusty plasmas." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Pulsed shear motion in a three-dimensional dusty plasma under microgravity conditions." Presented at the 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00004. http://meetings.aps.org/Meeting/DPP20/Session/CP12.4 , Nov-2020
Abstracts for Journals and Proceedings	Goree J, Liu B, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Correlation and spectrum of dust acoustic waves in a plasma under microgravity conditions." Presented at the 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00005. http://meetings.aps.org/Meeting/DPP20/Session/CP12.5 , Nov-2020
Articles in Peer-reviewed Journals	Goree J, Liu B, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Molotkov VI, Usachev AD, Petrov OF, Thoma MH, Thomas E Jr, Konopka U, Prokopiev S. "Correlation and spectrum of dust acoustic waves in a radio-frequency plasma using PK-4 on the International Space Station." Phys Plasmas. 2020 Dec;27(12):123701. https://doi.org/10.1063/5.0024500 , Dec-2020
Articles in Peer-reviewed Journals	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Petrov OF, Zobnin AV, Thoma MH. "Time-dependent shear motion in a strongly coupled dusty plasma in PK-4 on the International Space Station (ISS)." IEEE Transactions on Plasma Science. 2021 Sep;49:2972-78. http://dx.doi.org/10.1109/TPS.2021.3100300 , Sep-2021